



CSE490 (D) INTRODUCTION TO FOOTBALL DATA AND ANALYTICS



IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



- Professional Certificate from BFF
- Internship Opportunity at **BFF Data Science** Department after successful completion
- Learning from the **industry experts** of Football Data Science
- Opportunity to **co-author** research papers for **Sports Data Conferences**



CSE490 (D) Introduction to Football Data & Analytics Lecture-6

Prepared by-
Tarvir Anjum Aditto
Lecturer,
CSE, Brac University

Spreadsheet Scouting

- Scouting players based on certain attributes
- Sort from the best to worst
- **Not always guarantee success**
- Need to combine football, analytics and psychology.

Example sheet: [Soccer Stats In Excel - Premier League 2025-26 \(R01\) - Excel4Soccer](#)

Spreadsheet Scouting



- No. 1 in tackling and interception in league 1.
- Brought by leicester
- Leicester won the premier league.



- No. 2 in tackling and interception in league 1.
- Brought by Aston Villa
- Aston Villa was relegated

Both was undervalued before signing. Why the contrast in terms of result?

Scouting

- Recruitment often initiated by the agents!
- Clubs should choose a style of playing, set up their system, build algorithm to automatically identify candidates fit for the system.
- Player psychology & communication is important too!
- One optimal way:

Search using stat -> Send scout -> Analyze video data

Q. Can we find a way to calculate the individual contribution of a player?

The Naive Plus-Minus Statistic

Game 1: Player plays 60 minutes, team concedes 1 goal.

Game 2: Player plays 30 minutes, team scores 2 goals.

PM rating (on-pitch):

$$((-1 / 60) + (2 / 30)) \times 90 = +4.5$$

If the team lost the first game 1-0 and won the second 5-0, then:

PM rating (off-pitch):

$$((0 / 30) + (3 / 60)) \times 90 = +4.5$$

[\(PDF\) Plus-Minus Player Ratings for Soccer](#)

Limitation of the Naive Plus-Minus Statistic

- Ignores ratings of other players
- Does not account for opposition strength
- Measures only relative strength within the same team
- Does not account for match situation

Regularized Adjusted Plus Minus Statistic

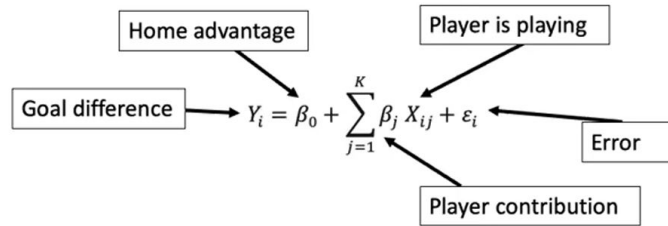
Dan Rosenbaum was likely the first to publish details of what is now known as an adjusted plus-minus rating, in an article posted on the web site 82games.com (Rosenbaum, 2004). His method involved splitting observations into two groups: **Home advantage** and **Player is playing**. He then ran the following multiple linear regression:

$$\text{Goal difference} \rightarrow Y_i = \beta_0 + \sum_{j=1}^K \beta_j X_{ij} + \varepsilon_i \quad (1)$$

Error

where the dependent variable Y_i is taken as the difference of home team points per possession and away team points per possession for the observation i . The independent variable X_{ij} is 1 if player j is playing at home, -1 if player j is playing away, and 0 otherwise. Then, β_0 measures the average home court advantage, and β_j for $j = 1, \dots, K$ measures the difference between player j and a set of reference players. This set of reference players refers to all players with less

Regularized Adjusted Plus Minus Statistic



Minutes	Y_i	X_{i1} (home)	X_{i2} (home)	X_{i3} (away)	X_{i4} (away)
0-60	2/60	1	1	0	1
61-68	0/8	1	1	1	0
69-85	-1/16	0	1	1	0

Player 4->3

Player 5->1

Regularized Adjusted Plus Minus Statistic

regularization and cross validation to improve the ratings. Using the method of ordinary least squares, the parameters of model (1) would be found by minimizing

$$\sum_i \left(Y_i - \beta_0 - \sum_{j=1}^K \beta_j X_{ij} \right)^2 \quad (3)$$

A Bayesian technique of regularization, known as ridge regression or Tikhonov regularization, implies that one instead minimizes

$$\sum_i \left(Y_i - \beta_0 - \sum_{j=1}^K \beta_j X_{ij} \right)^2 + \lambda \sum_{j=0}^K (\beta_j)^2$$

This corresponds to using a Gaussian prior distribution, stabilizing the estimated ratings, both in cases with few observations and when multicollinearity is present (Thomas et al., 2013). Cross

Regularized Adjusted Plus Minus Statistic

$$\sum_i \left(Y_i - \beta_0 - \sum_{j=1}^K \beta_j X_{ij} \right)^2 + \lambda \sum_{j=0}^K (\beta_j)^2$$

Usual maximum
likelihood
minimisation.

Punishment for
too many
paramters. Most
players set to
zero.

The EA Sports Player Performance Index

Performed a linear regression on how the number of particular actions impacted the number of goals. Below is the regression output for the model –

Variables	Coef.	Std. err.	t-stat	p-value
Crosses	0.519	0.069	7.490	0.000
Dribbles	0.118	0.026	4.600	0.000
Passes	0.034	0.004	7.830	0.000
Opposition interceptions	-0.024	0.009	-2.520	0.012
Opposition yellows	0.253	0.134	1.890	0.059
Opposition reds	1.023	0.506	2.020	0.043
Opposition tackle win ratio	-0.170	0.090	-1.890	0.059
Opposition cleared	-0.017	0.009	-1.920	0.055
Constant	6.463	0.815	7.930	0.000

[On the Development of a Soccer Player Performance Rating System for the English Premier League | Interfaces](#)

The EA Sports Player Performance Index

- Built a statistical model linking player actions to goal creation.
- Counted actions per match:
 - Passing. Tackling
 - Crossing
 - Dribbling
 - Blocking & clearing
 - Yellow/red cards
- Used statistical fitting to estimate how these actions predict:
 - Team shots
 - Opponent shots

Key Findings (Premier League 2003–2006):

- More passes → more chances created
- Successful crosses are highly valuable
- 1 successful cross ≈ 10 standard passes (in shot creation value)
- A red card ≈ losing the value of 41 interceptions

The EA Sports Player Performance Index

Who Was “Best” According to the Model?

- Top player: **Mark Schwarzer (Fulham, GK)** = 7.29
- 2nd: Gareth Barry = 7.06
- 3rd: Sol Campbell = 6.86
- No Manchester United player in top 20
- Only one striker (Nicolas Anelka) and one midfielder (Barry) in top ranks
- Majority were **goalkeepers and defenders**

Why Defenders Ranked Higher

- Strikers & attacking midfielders were inefficient finishers
- Example: Frank Lampard => highly variable shooting
- Defensive actions (interceptions, blocks) prevented goals
- Preventing goals > scoring goals (in model value)

Name	Team	Position	Subindex 1 points
Mark Schwarzer	Fulham	Goalkeeper	7.29
Gareth Barry	Aston Villa	Midfielder	7.06
Sol Campbell	Portsmouth	Defender	6.86
Gary Cahill	Bolton Wanderers	Defender	6.70
J. Lloyd Samuel	Bolton Wanderers	Defender	6.63
Robert Green	West Ham United	Goalkeeper	6.45
Heurelho Gomes	Tottenham Hotspur	Goalkeeper	6.43
David James	Portsmouth	Goalkeeper	6.40
Scott Carson	West Bromwich Albion	Goalkeeper	6.39
Nicolas Anelka	Chelsea	Striker	6.33
Richard Dunne	Manchester City	Defender	6.30
Andrew O'Brien	Bolton Wanderers	Defender	6.26
Jussi Jaaskelainen	Bolton Wanderers	Goalkeeper	6.20
Sylvain Distin	Portsmouth	Defender	6.12
Thomas Sorensen	Stoke City	Goalkeeper	6.10
Matthew Upson	West Ham United	Defender	6.03
Ryan Shawcross	Stoke City	Defender	5.85
Danny Collins	Sunderland	Defender	5.81
Michael Turner	Hull City	Defender	5.54
Antony Ferdinand	Sunderland	Defender	5.47

The EA Sports Player Performance Index

Name	Team	Position	Index score
Nicolas Anelka	Chelsea	Striker	751
Dirk Kuyt	Liverpool	Midfielder	655
Frank Lampard	Chelsea	Midfielder	653
Dimitar Berbatov	Manchester United	Striker	586
Gabriel Agbonlahor	Aston Villa	Striker	575
Jose Reina	Liverpool	Goalkeeper	573
Gareth Barry	Aston Villa	Midfielder	564
Jamie Carragher	Liverpool	Defender	552
Steven Gerrard	Liverpool	Midfielder	540
Nemanja Vidic	Manchester United	Defender	540
Cristiano Ronaldo	Manchester United	Striker	517
Petr Cech	Chelsea	Goalkeeper	513
John Terry	Chelsea	Defender	501
Robbie Keane	Tottenham Hotspur	Striker	482
Edwin Van der Sar	Manchester United	Goalkeeper	481
Neves Denilson	Arsenal	Midfielder	477
Mark Schwarzer	Fulham	Goalkeeper	477
Joleon Lescott	Everton	Defender	476
Tim Howard	Everton	Goalkeeper	472
Kevin Davies	Bolton Wanderers	Striker	468

Can be improved by adding goals,
clean sheets, wins, etc.

But this is somewhat unprincipled.